

Lab Report No 7
**Implementation of autocorrelation/cross
correlation**
Digital System Processing



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Lab Example No 1:

Solution:

Brief description (3-5 lines)

Auto Correlation Function:

Auto correlation function is a measure of similarity between a signal & its time delayed version. It is represented with $R(k)$.

The auto correlation function of $x(n)$ is given by

$R_{11}(k)=R(k)=\sum_{n=-\infty}^{\infty} x_1(n)x_2(n-k)$

Cross-correlation Function:

Cross-correlation is a measure of similarity between two signals as a function of the time delay applied to one of the signals. It is commonly used in signal processing and communication systems to identify time delays and patterns in signals. In this section, we will discuss the formal theory and provide the Python code to compute the cross-correlation of two signals. The cross-correlation between two discrete-time signals $x(n)$ and $y(n)$ is defined as:
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$R_{xy}(m) = \sum_n [x(n) * y(n+m)]$

where m is the time shift, and $R_{xy}(m)$ is the cross-correlation of x and y at time shift m .
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The code

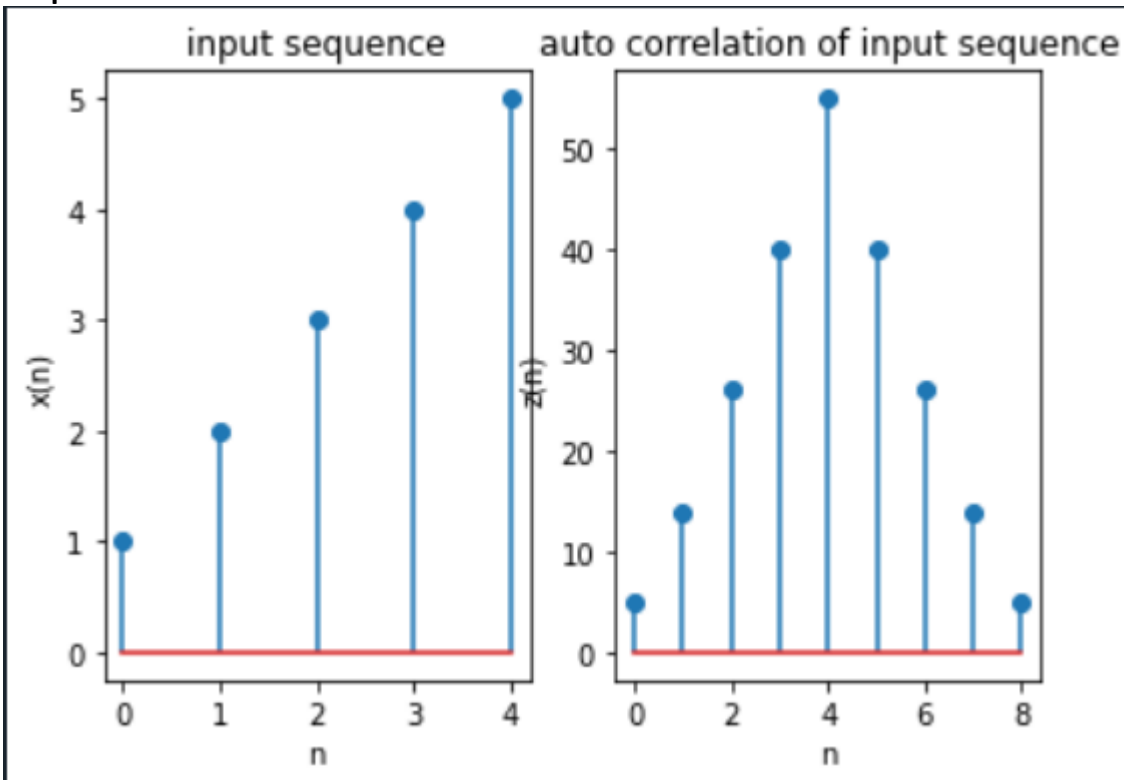
```

1  # autocorrelation/cross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  x = np.array([1, 2, 3, 4, 5])
5  plt.subplot(1, 2, 1)
6  plt.stem(x)
7  plt.xlabel('n')
8  plt.ylabel('x(n)')
9  plt.title('input sequence')
10 # Auto-correlation of input sequence
11 z = np.correlate(x, x, mode='full')
12 print('The values of z are = ')
13 print(z)
14 plt.subplot(1, 2, 2)
15 plt.stem(z)
16 plt.xlabel('n')
17 plt.ylabel('z(n)')
18 plt.title('auto correlation of input sequence')
19 plt.show()

```

The results (Screenshot)

Graph:



OUTPUT:

```
The values of z are =  
[ 5 14 26 40 55 40 26 14 5]
```

Lab Example No 2:

Solution:

Brief description (3-5 lines)

Auto Correlation Function:

Auto correlation function is a measure of similarity between a signal & its time delayed version. It is represented with $R(k)$.

The auto correlation function of $x(n)$ is given by

$$R_{11}(k) = R(k) = \sum_{n=-\infty}^{\infty} x_1(n)x_2(n-k)$$

Cross-correlation Function:

Cross-correlation is a measure of similarity between two signals as a function of the time delay applied to one of the signals. It is commonly used in signal processing and communication systems to identify time delays and patterns in signals. In this section, we will discuss the formal theory and provide the Python code to compute the cross-correlation of two signals. The cross-correlation between two discrete-time signals $x(n)$ and $y(n)$ is defined as:

$$R_{xy}(m) = \sum_n [x(n) * y(n+m)]$$

where m is the time shift, and $R_{xy}(m)$ is the cross-correlation of x and y at time shift m .

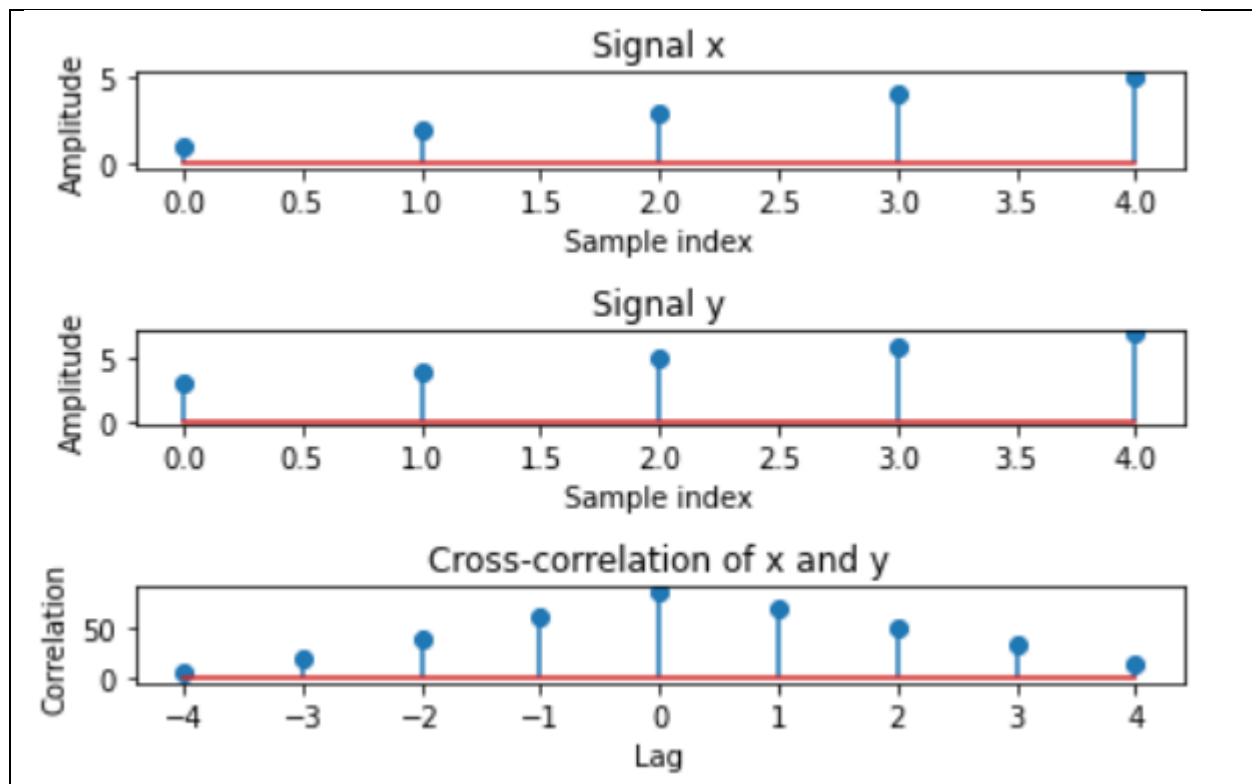
The code

```

1  # autocorrelation and cross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  x = np.array([1, 2, 3, 4, 5])
5  y = np.array([3, 4, 5, 6, 7])
6  r_xy = np.correlate(x, y, mode='full')
7  range = np.arange(-len(x) + 1, len(x))
8  plt.subplot(3, 1, 1)
9  plt.stem(x)
10 plt.xlabel('Sample index')
11 plt.ylabel('Amplitude')
12 plt.title('Signal x')
13 plt.subplot(3, 1, 2)
14 plt.stem(y)
15 plt.xlabel('Sample index')
16 plt.ylabel('Amplitude')
17 plt.title('Signal y')
18 plt.subplot(3, 1, 3)
19 plt.stem(range, r_xy)
20 plt.xlabel('Lag')
21 plt.ylabel('Correlation')
22 plt.title('Cross-correlation of x and y')
23 plt.tight_layout()
24 plt.show()

```

The results (Screenshot)



Lab Task No 1:

Solution:

Brief description (3-5 lines)

Find the cross correlation of the sequences
 $x(n) = \{3, 5, 1, 2\}$ and $h(n) = \{1, 4, 3\}$

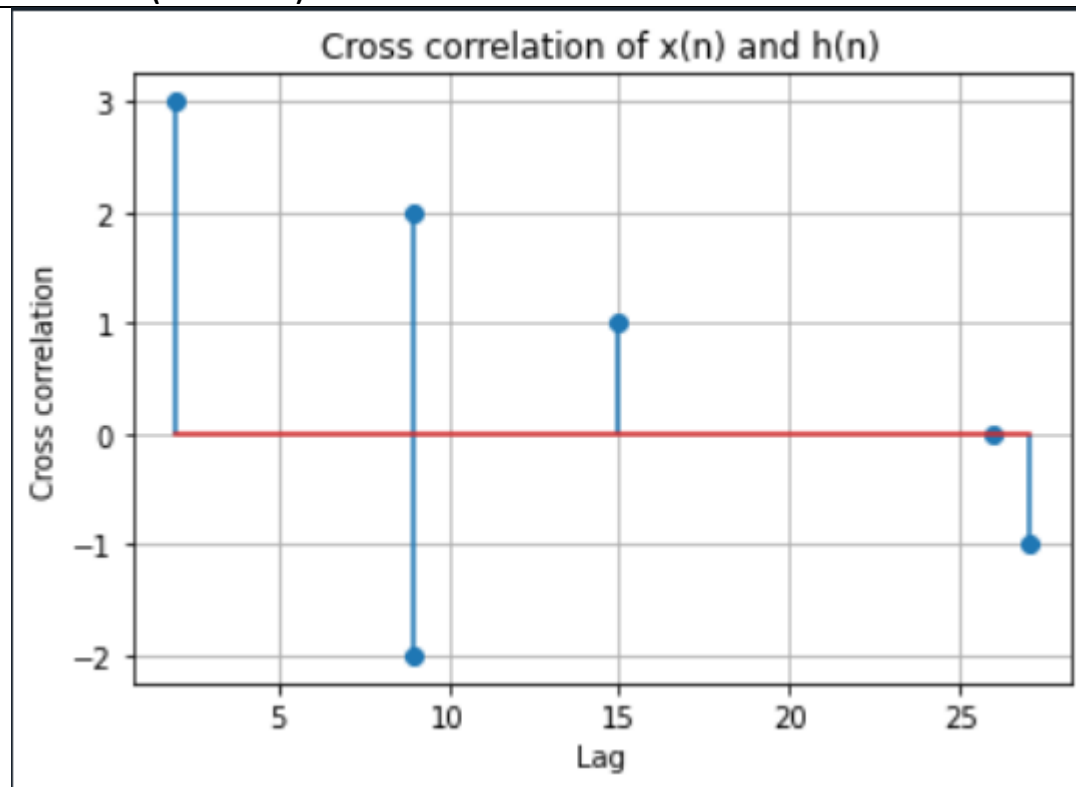
The code

```

1 # autocorrelationandcross correlation
2 import numpy as np
3 import matplotlib.pyplot as plt
4 x=np.array([3,5,1,2])
5 h=np.array([1,4,3])
6 corr=np.convolve(x,h[::-1],mode='full')
7 lags=np.arange(-len(h)+1,len(x))
8 plt.stem(corr,lags)
9 plt.xlabel('Lag')
10 plt.ylabel('Cross correlation')
11 plt.title('Cross correlation of x(n) and h(n)')
12 plt.grid()
13 plt.show()

```

The results (Screenshot)



Lab Task No 2:

Solution:

Brief description (3-5 lines)

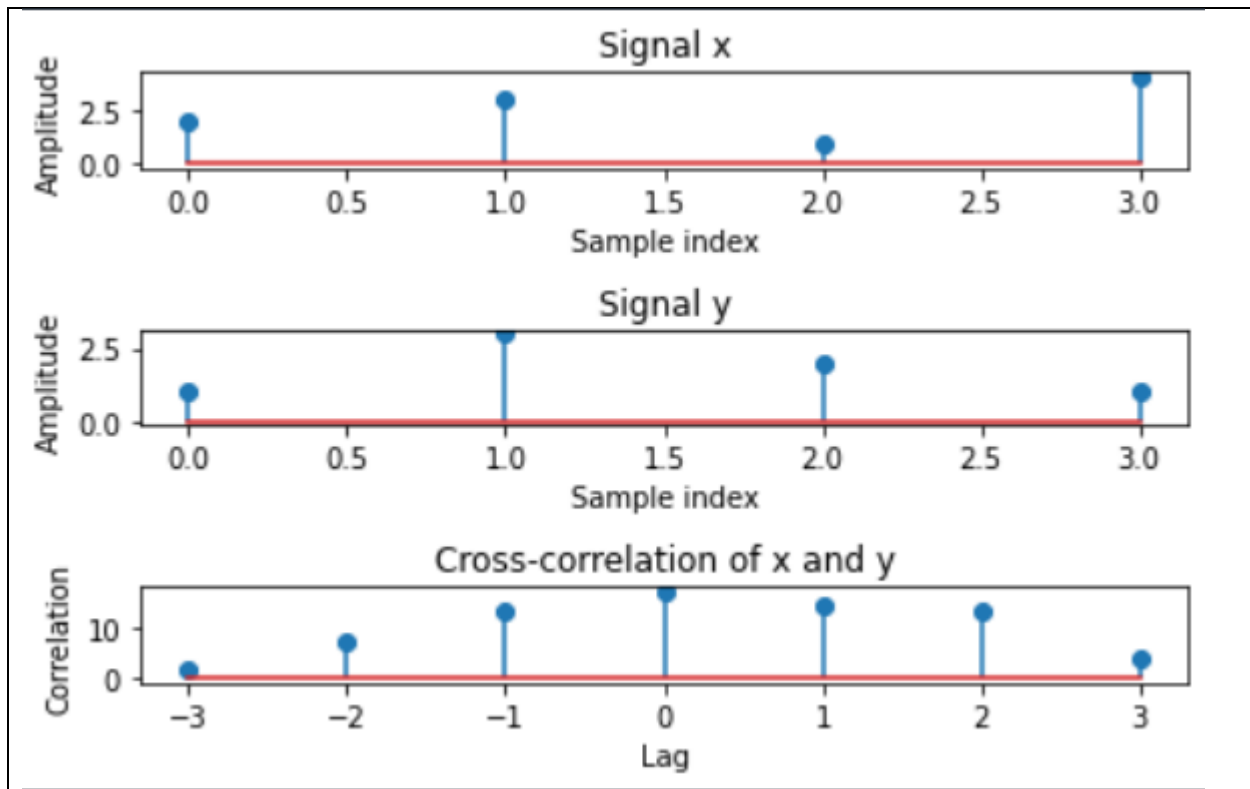
Find the cross correlation of the sequences

$x(n)=\{2,3,1,4\}$ and $y(n)=\{1,3,2,1\}$

The code

```
1  # autocorrelation and cross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  x = np.array([2,3,1,4])
5  y = np.array([1,3,2,1])
6  r_xy = np.correlate(x, y, mode='full')
7  range = np.arange(-len(x) + 1, len(x))
8  plt.subplot(3, 1, 1)
9  plt.stem(x)
10 plt.xlabel('Sample index')
11 plt.ylabel('Amplitude')
12 plt.title('Signal x')
13 plt.subplot(3, 1, 2)
14 plt.stem(y)
15 plt.xlabel('Sample index')
16 plt.ylabel('Amplitude')
17 plt.title('Signal y')
18 plt.subplot(3, 1, 3)
19 plt.stem(range, r_xy)
20 plt.xlabel('Lag')
21 plt.ylabel('Correlation')
22 plt.title('Cross-correlation of x and y')
23 plt.tight_layout()
24 plt.show()
```

The results (Screenshot)



Lab Task No 3(a):

Solution:

Brief description (3-5 lines)

Find the autocorrelation of the sequences
 $x(n) = \{1, 3, 0, 4\}$

The code

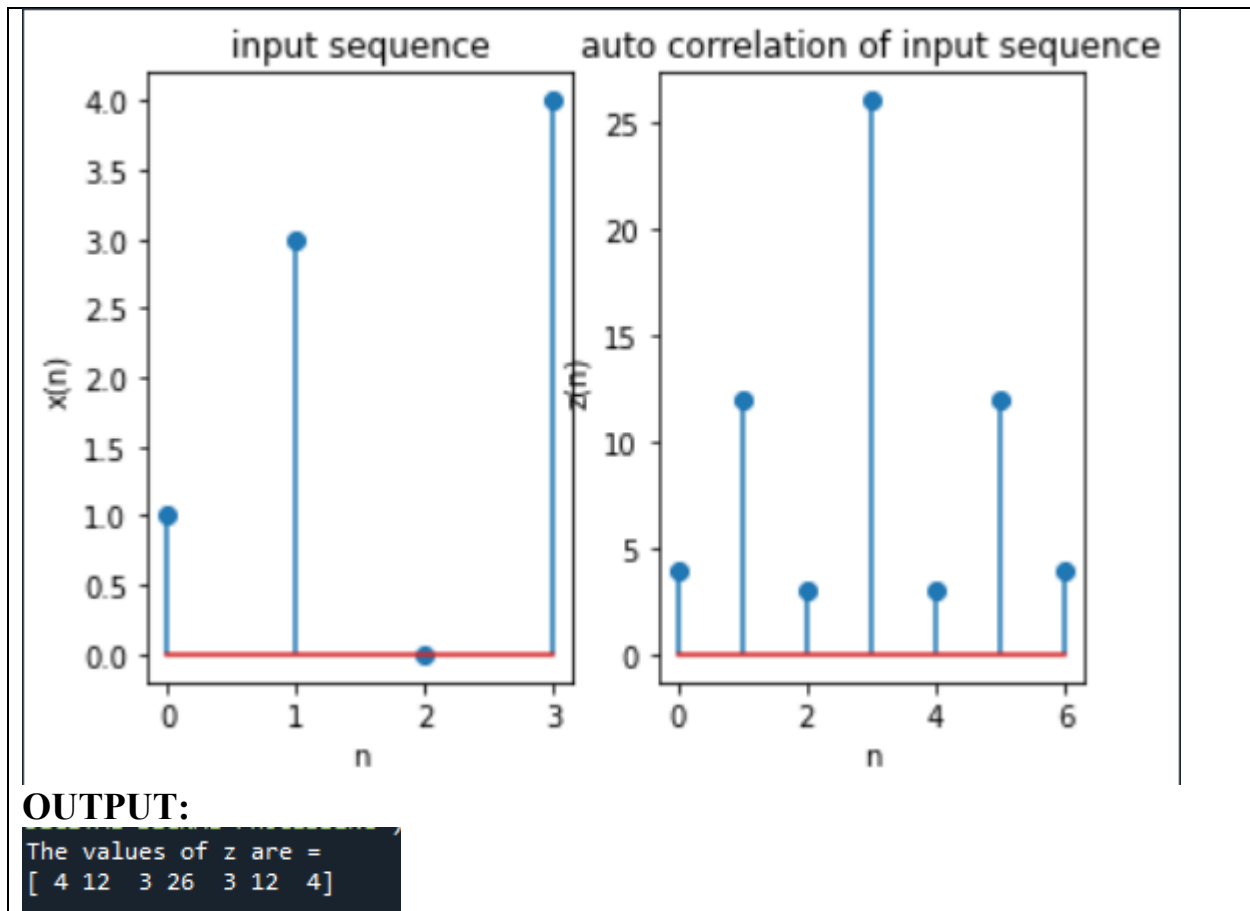
```

1  # autocorrelationandcross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  x = np.array([1,3,0,4])
5  plt.subplot(1,2,1)
6  plt.stem(x)
7  plt.xlabel('n')
8  plt.ylabel('x(n)')
9  plt.title('input sequence')
10 # Auto-correlation of input sequence
11 z = np.correlate(x, x, mode='full')
12 print('The values of z are = ')
13 print(z)
14 plt.subplot(1, 2, 2)
15 plt.stem(z)
16 plt.xlabel('n')
17 plt.ylabel('z(n)')
18 plt.title('auto correlation of input sequence')
19 plt.show()

```

The results (Screenshot)

Graph:



Lab Task No 3(b):

Solution:

Brief description (3-5 lines)

Find the autocorrelation of the sequences

$$y(n) = \{2, 1, -2, 1\}$$

The code

```

1  # autocorrelation and cross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  y= np.array([2,1,-2,1])
5  plt.subplot(1,2,1)
6  plt.stem(y)
7  plt.xlabel('n')
8  plt.ylabel('y(n)')
9  plt.title('input sequence')
10 z = np.correlate(y, y, mode='full')
11 print('The values of z are = ')
12 print(z)
13 plt.subplot(1, 2, 2)
14 plt.stem(z)
15 plt.xlabel('n')
16 plt.ylabel('z(n)')
17 plt.title('auto correlation of input sequence')
18 plt.show()

```

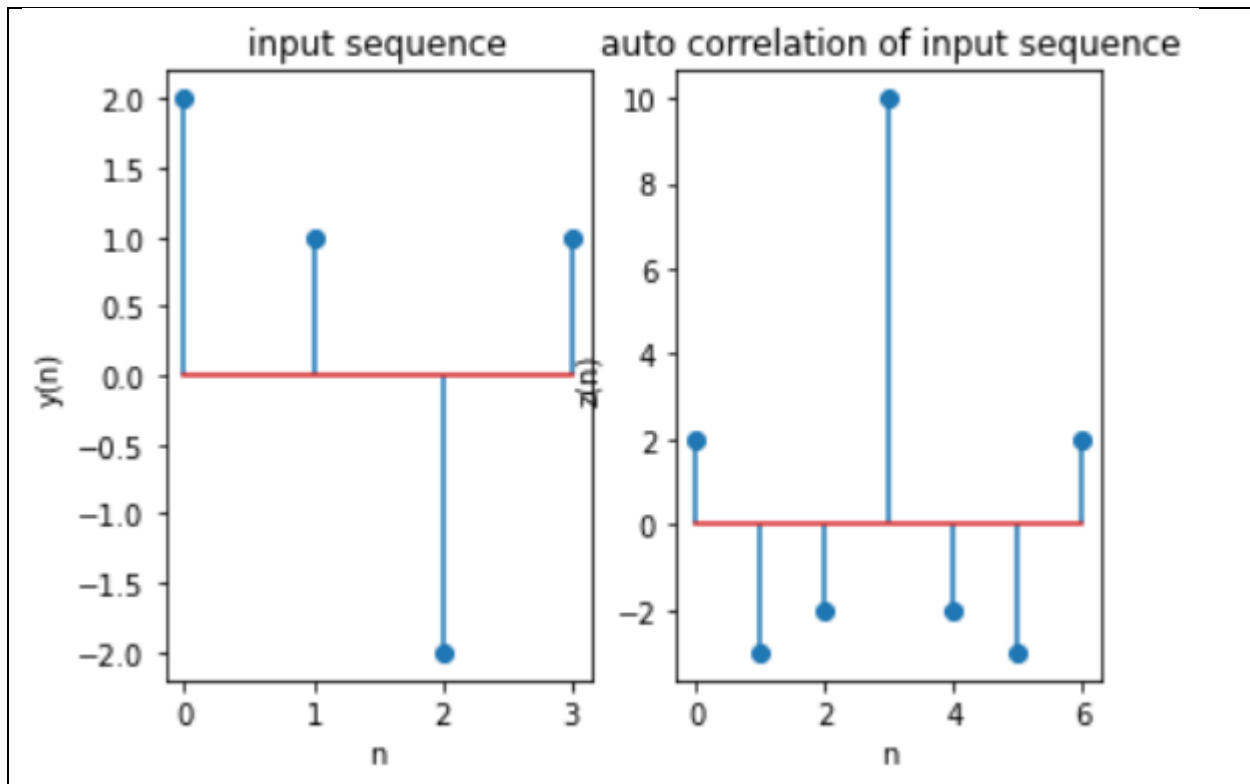
The results (Screenshot)

```

The values of z are =
[ 2 -3 -2 10 -2 -3  2]

```

Graph:



Lab Task No 2:

Solution:

Brief description (3-5 lines)

Find the discrete periodic cross correlation $R_{xy}(n)$ and $R_{yx}(n)$ of the whose first period is given by:

$x(n) = \{1, 3, 0, 4\}$ and $y(n) = \{2, 1, -2, 1\}$

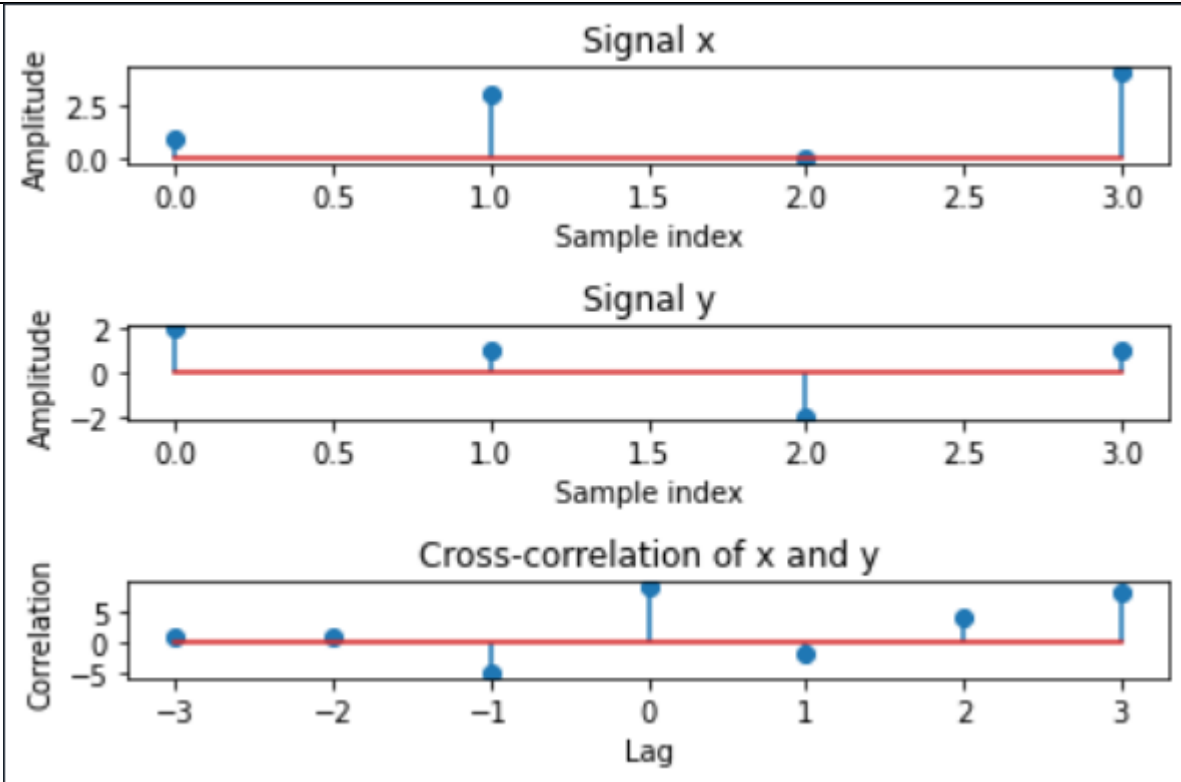
The code

```

1  #autocorrelationandcross correlation
2  import numpy as np
3  import matplotlib.pyplot as plt
4  x = np.array([1,3,0,4])
5  y = np.array([2,1,-2,1])
6  r_xy = np.correlate(x, y, mode='full')
7  range = np.arange(-len(x) + 1, len(x))
8  plt.subplot(3, 1, 1)
9  plt.stem(x)
10 plt.xlabel('Sample index')
11 plt.ylabel('Amplitude')
12 plt.title('Signal x')
13 plt.subplot(3, 1, 2)
14 plt.stem(y)
15 plt.xlabel('Sample index')
16 plt.ylabel('Amplitude')
17 plt.title('Signal y')
18 plt.subplot(3, 1, 3)
19 plt.stem(range, r_xy)
20 plt.xlabel('Lag')
21 plt.ylabel('Correlation')
22 plt.title('Cross-correlation of x and y')
23 plt.tight_layout()
24 plt.show()

```

The results (Screenshot)



Conclusion

In conclusion, Autocorrelation measures a signal's similarity to itself over time, aiding in identifying periodic patterns or dependencies within data. Cross-correlation compares two signals to find similarities or correlations between them, useful for identifying relationships between different data sets or signals. Both techniques are valuable in signal processing, time series analysis, and various fields where understanding temporal relationships is crucial.